

EEG/ERP Preprocessing Manual

A practical workflow using EEGLAB and ERPLAB

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Contents

1. Basic Workflow	3
1.1 Single-participant preprocessing workflow	3
1.2 Batch processing	4
2. Identifying Bad Segments and Bad Channels	5
2.1 Characteristics of Good-Quality EEG Signals	5
2.2 Ocular artifact	6
2.3 Muscle artifact.....	7
2.4 Movement artifacts.....	8
2.5 Bad electrodes	9
2.6 Recording discontinuities and electrode pops	10
2.7 Slow baseline drift.....	11
Resources	12
3. Identifying Artifactual Independent Components	13
3.1 Evaluate multiple views together.....	13
3.2 Brain-related components	13
3.3 Ocular components.....	14
3.4 Muscle components	15
3.5 Cardiac components	17
3.6 Line-noise components	17
3.7 Channel-noise components	17
Resources	18
4. Special Processing: ICA-Weight Transfer.....	19
4.1 Rationale for using separate ERP-analysis and ICA-training branches	19
4.2 Requirements for valid transfer across branches	19
4.3 Recommended processing sequence.....	20
4.4 Use of ICLabel across branches	20
5. Reference.....	22
6. Online Resources	23

1. Basic Workflow

1.1 Single-participant preprocessing workflow

Step	Processing Step	Description	EEGLAB/ERPLAB GUI Path
1	Import data	Import the dataset to be preprocessed or analyzed. <i>.set</i> is EEGLAB's native dataset format. Other file formats (e.g., <i>.cdt</i>) may require additional EEGLAB plug-ins for import.	File → Import data → Using EEGLAB functions and plug-ins; or File → Load existing dataset
2	Remove non-analysis channels	Remove or exclude channels that will not be included in scalp-EEG analysis, such as TRIGGER and VEOG channels.	Edit → Select data → Channels/channel types. Select the On/Off checkbox as appropriate to invert the channel selection.
3	Load channel locations	Load the appropriate electrode-location file and save a complete channel-location template for subsequent processing.	Edit → Channel locations → Read locations
4	Filter the data	Apply the project-defined high-pass and low-pass filters to the continuous EEG data. If a notch filter is used to suppress line noise, select the appropriate frequency according to the recording location. The mains frequency is typically 50 Hz in mainland China and Hong Kong (e.g., a 48–52 Hz notch), and 60 Hz in the United States, Canada, and Taiwan. Japan uses either 50 or 60 Hz depending on the region.	Tools → Filter the data → Basic FIR filter
5	Remove bad segments and bad channels	Remove non-stereotyped extreme artifacts, recording dropouts, amplifier saturation, and prolonged periods of high-amplitude movement artifact before ICA; do not remove stereotyped blinks simply because ICA can often isolate them. Identify channels that are persistently noisy, flat, or poorly connected.	Tools → Inspect/reject data by eye
6	Re-reference the data	Apply the reference scheme specified in the study protocol, such as the average of the bilateral mastoids or the average reference across all scalp EEG channels.	Tools → Re-reference the data
7	Run ICA	Perform independent component analysis (ICA) on the preprocessed EEG data to obtain independent components for subsequent artifact identification and removal.	Tools → Decompose data by ICA
8	Identify and remove artifact-related ICs	Evaluate the scalp topography, component time series, power spectrum, and ERP image jointly when classifying independent components. Use ICLabel as decision support, followed by manual inspection of components classified as artifacts or otherwise ambiguous.	View ICLabel results: Tools → Classify components using ICLabel. Inspect/label ICs: Tools → Inspect/label components by map
9	Extract bin-based epochs	Create and archive the Eventlist, prepare the bin descriptor file, run BINLISTER to assign events to bins, and extract bin-based epochs.	ERPLAB → EventList → Create EEG EventList ERPLAB → Assign Bins

Step	Processing Step	Description	EEGLAB/ERPLAB GUI Path
			(BINLISTER)
			ERPLAB → Extract Bin-Based Epochs
10	Apply baseline correction	Apply baseline correction to each epoch using the project-defined baseline interval.	Tools → Remove epoch baseline
11	Remove residual extreme artifacts	Apply the project-defined artifact-rejection procedures, such as absolute-voltage thresholds, moving-window peak-to-peak thresholds, step-like artifact detection, or other predefined criteria. Manually review the marked epochs before ERP averaging.	ERPLAB → Artifact detection in epoched data
12	Average	Compute the ERPs	ERPLAB → Compute averaged ERP

Note: These preprocessing steps can be broadly conceptualized as two types: **parameter-driven steps** and **judgment-based steps**. Parameter-driven steps are determined primarily by predefined parameters and analysis settings, whereas judgment-based steps rely more heavily on the researcher's visual inspection, domain knowledge, and practical experience.

Decision Type	Examples
Parameter-driven	Filter cutoffs; epoch window; baseline interval; voltage thresholds; reference scheme; ICA algorithm
Judgment-based	Identification of bad segments; identification of bad channels; classification of ambiguous ICs

1.2 Batch processing

Batch processing using scripts can substantially improve preprocessing efficiency, ensure consistency across participants, and facilitate data management. A complete example of a batch-processing script used in one of my projects is provided in **Attachment 2**.

2. Identifying Bad Segments and Bad Channels

Being able to identify bad segments and bad channels, as well as understand their likely causes, is useful in two situations:

During data acquisition: to recognize abnormal EEG patterns during online monitoring and take appropriate action when necessary.

During preprocessing: to determine which segments or channels should be removed or excluded before ICA and subsequent preprocessing steps.

General criteria:

- Do not remove stereotyped blink artifacts simply because they have large amplitudes; ICA can often isolate these artifacts effectively.
- Remove unique, non-repetitive artifacts that may consume ICA dimensions, such as electrode pops, amplifier saturation, or gross movement artifacts.
- Document channels excluded before ICA and interpolate them at the appropriate stage of preprocessing.

2.1 Characteristics of Good-Quality EEG Signals

A good-quality EEG recording should show continuous and physiologically plausible fluctuations across channels, without persistent high-amplitude noise, abrupt voltage jumps, prolonged flat signals, or obvious recording dropouts. The waveforms should generally appear smooth and stable over time, although normal EEG activity will naturally vary across channels and time. Neighboring scalp electrodes often show broadly similar activity patterns because of volume conduction, but the signals should not be completely identical.

When visually inspecting the data, pay particular attention to the following features:

- **Continuous signal:** no abrupt interruptions, missing data, or prolonged flat periods.
- **Stable amplitude:** no persistent extreme amplitudes or sudden large voltage jumps.
- **Reasonable spatial consistency:** neighboring electrodes usually show related, but not identical, activity.
- **Limited high-frequency contamination:** the signal should not be dominated by continuous “fuzzy” or dense high-frequency activity.
- **No persistent channel-specific abnormality:** a single channel should not remain substantially noisier, flatter, or more unstable than the surrounding channels.
- **Physiological variability:** normal EEG is not perfectly smooth or regular; some temporal and spatial variation is expected.

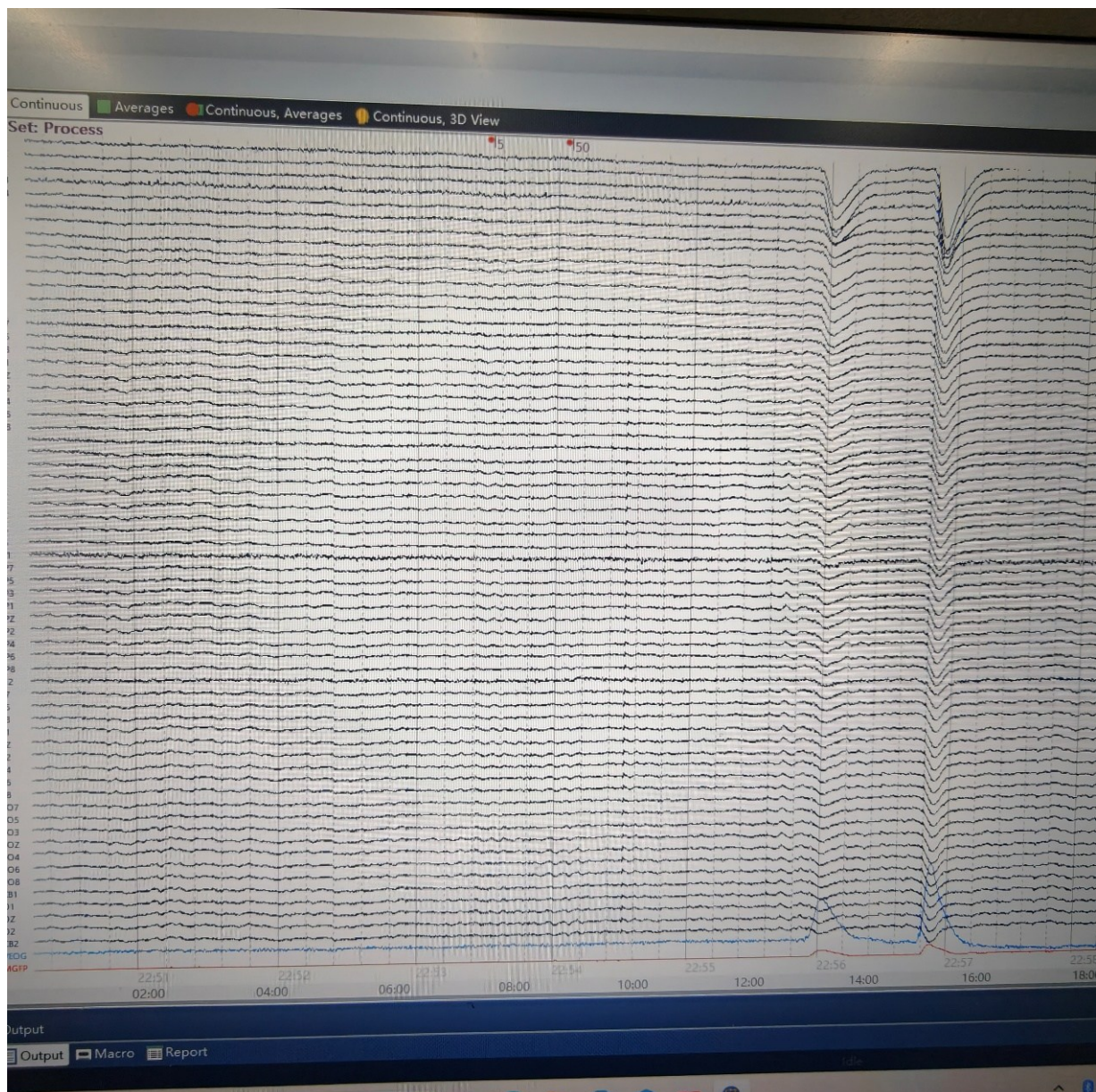


Figure 1. Example of relatively clean continuous EEG with stable channel amplitudes and no obvious sustained artifact.

2.2 Ocular artifact

Eye blinks and lateral eye movements are usually largest at frontal channels. Blinks often produce broad, stereotyped deflections; horizontal eye movements often show **opposite polarity** over left and right frontal sites. These patterns may be retained for ICA if they are not accompanied by gross movement or amplifier saturation.

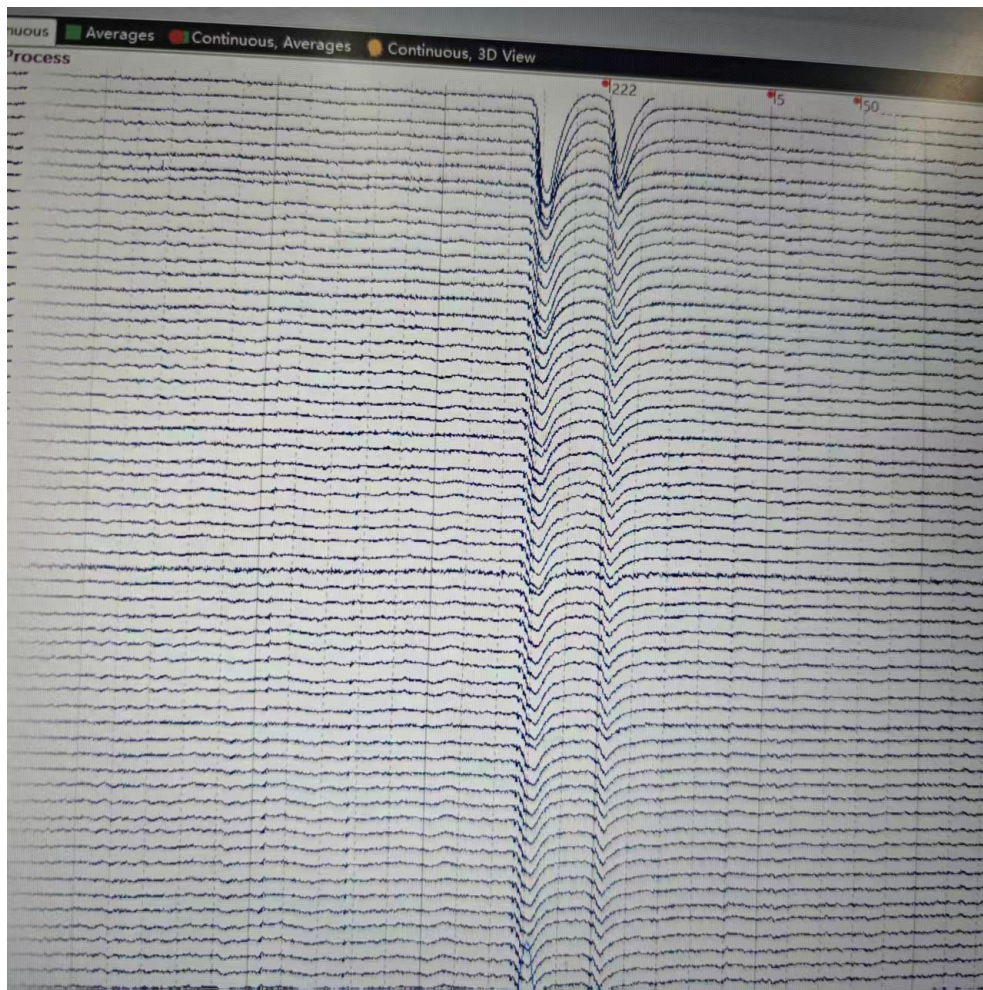


Figure 2. Example of a large ocular deflection visible across channels, with strongest expression at anterior sites.

2.3 Muscle artifact

Muscle activity typically appears as irregular, relatively high-frequency EEG activity and is often spatially focal. Muscle artifacts are commonly prominent over temporal and peripheral scalp electrodes because of their proximity to cranial and facial muscles. For example, activity related to muscles around the ears and temporal region may be particularly visible at electrodes such as TP7 and TP8 (Figure 3).

Muscle artifacts may also appear over frontal electrodes. For example, excessive pressure from a tight-fitting EEG cap may cause discomfort or sustained contraction of the forehead muscles, resulting in prominent high-frequency activity over frontal channels (Figure 4).

If muscle artifacts are observed during recording, adjust the EEG cap when appropriate and safe, and remind the participant to relax, minimize unnecessary body movements, and avoid clenching the jaw or teeth.

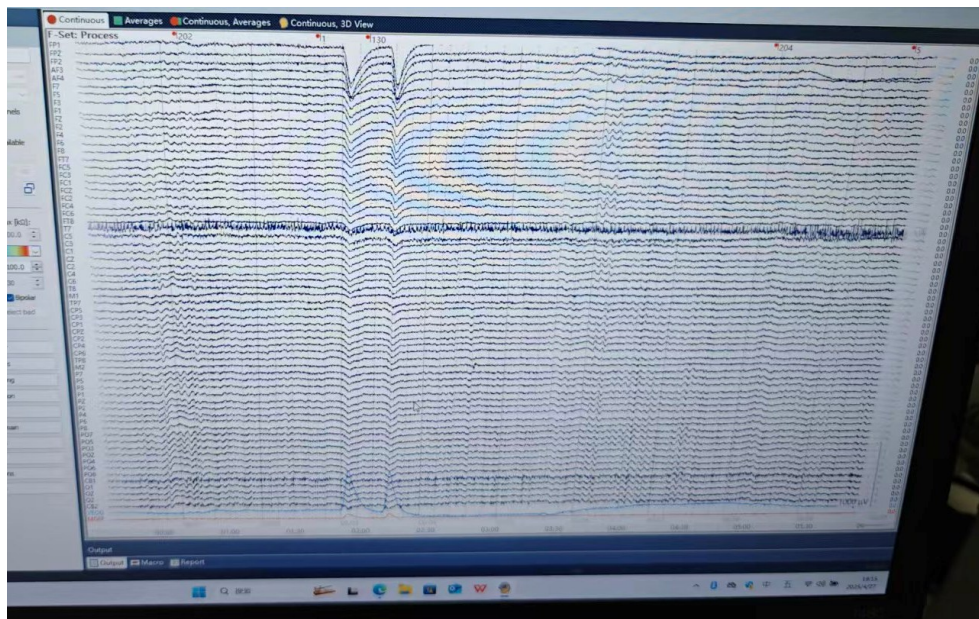


Figure 3. Peri-auricular muscle activity, often prominent near temporal electrodes such as TP7 and TP8.

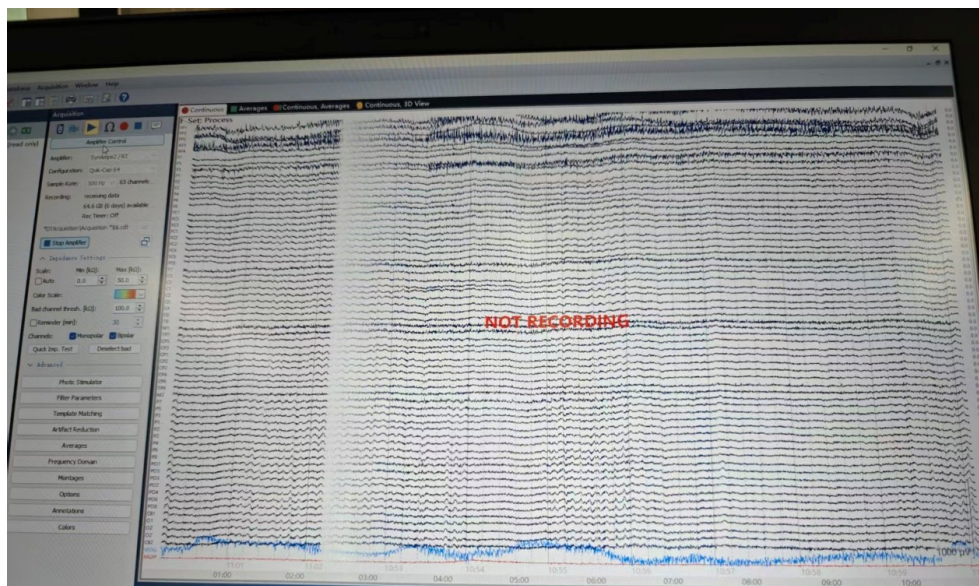


Figure 4. Pronounced frontal muscle activity; an overly tight cap may contribute.

2.4 Movement artifacts

Gross participant movements can produce large, widespread, and irregular deflections across multiple EEG channels. These artifacts may result from a combination of head or body movement, electrode displacement, changes in electrode–skin contact, cable movement, and accompanying muscle activity.

Possible causes include:

- Sudden changes in sitting posture or body position.
- Head turning, nodding, or other large head movements.
- Movement of the EEG cap, electrodes, or electrode cables.
- Abrupt changes in electrode–skin contact caused by movement.
- Concurrent contraction of facial, neck, or shoulder muscles.

If movement artifacts are observed, remind the participant to remain as still and relaxed as possible and to avoid unnecessary head or body movements. Check whether the electrode cap or cables have shifted and readjust them when appropriate and safe.

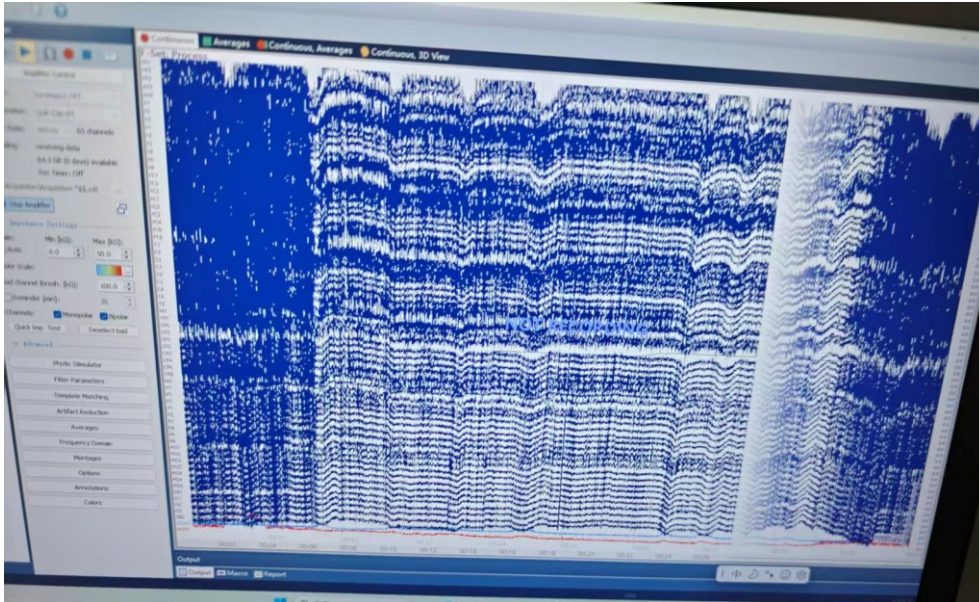


Figure 5. Gross movement affecting many channels.

2.5 Bad electrodes

A channel should be marked as bad only when the abnormality is persistent or recurrent across a substantial portion of the recording. A brief artifact at one electrode is not sufficient by itself.

- Common indicators: prolonged flatline, repeated electrode pops, excessive broadband noise, persistent line noise, unusually large variance, or poor correlation with neighboring channels.
- Spatial clue: the abnormality remains strongly localized to one electrode while nearby electrodes show plausible EEG.
- Handling: remove the channel before ICA, save its label, and interpolate it after IC rejection using the original montage.
- Exclusion: define a project-level rule for the maximum acceptable number or proportion of bad scalp channels.

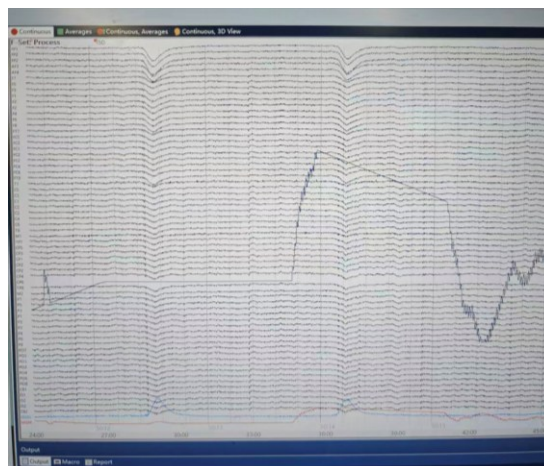


Figure 6. Example of a persistently abnormal channel that should be removed before ICA and interpolated later.

2.6 Recording discontinuities and electrode pops

Recording discontinuities and electrode pops are characterized by abrupt, non-physiological changes in the EEG signal. They may appear as sudden voltage jumps, sharp transients, step-like shifts, or brief disruptions in otherwise continuous EEG activity.

Possible causes include:

- **Electrode pops or abrupt changes in electrode–skin contact:** Sudden changes in electrode contact or impedance can produce large step-like voltage shifts. These artifacts may affect a single channel or a small group of channels.
- **Reference-electrode disturbance:** Because multiple EEG channels share the same reference, a sudden change in the reference-electrode contact can produce widespread, simultaneous transients across many or nearly all channels.
- **Cable or connector movement:** Movement of electrode cables or loose connectors may introduce abrupt voltage changes or intermittent signal instability.

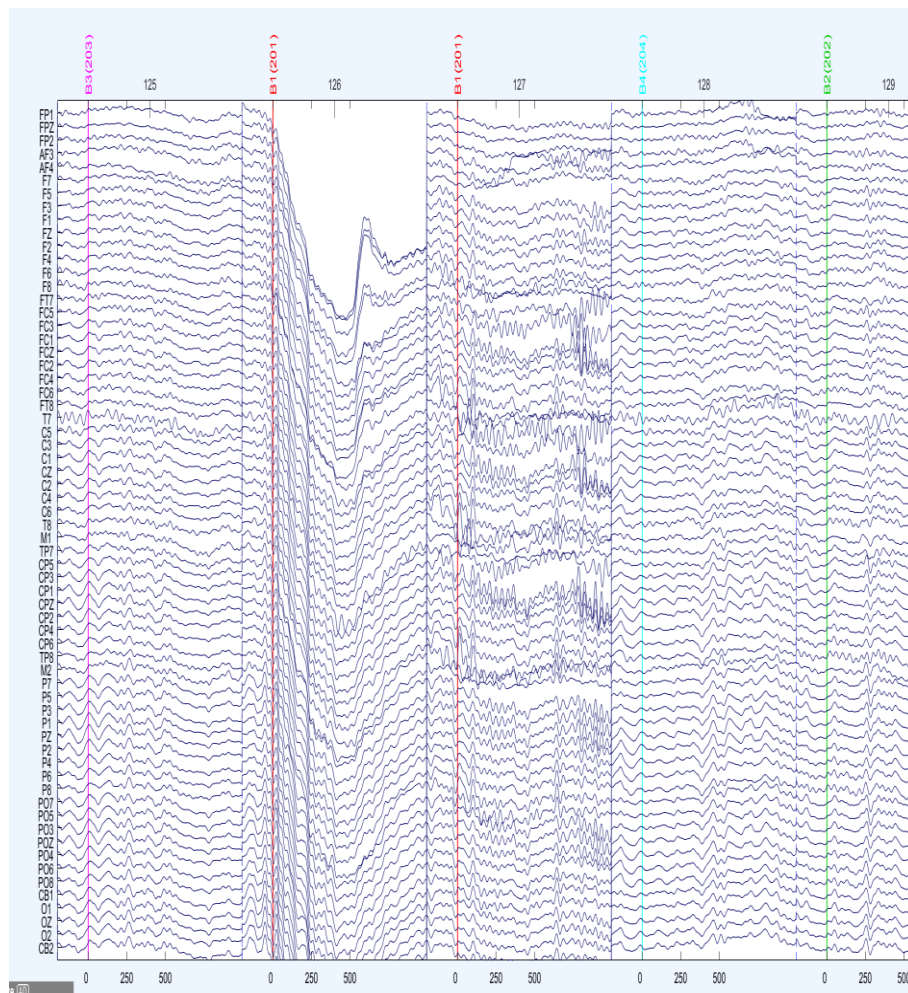


Figure 7. Example of a large step-like artifact, possibly caused by an electrode pop or abrupt change in electrode contact.

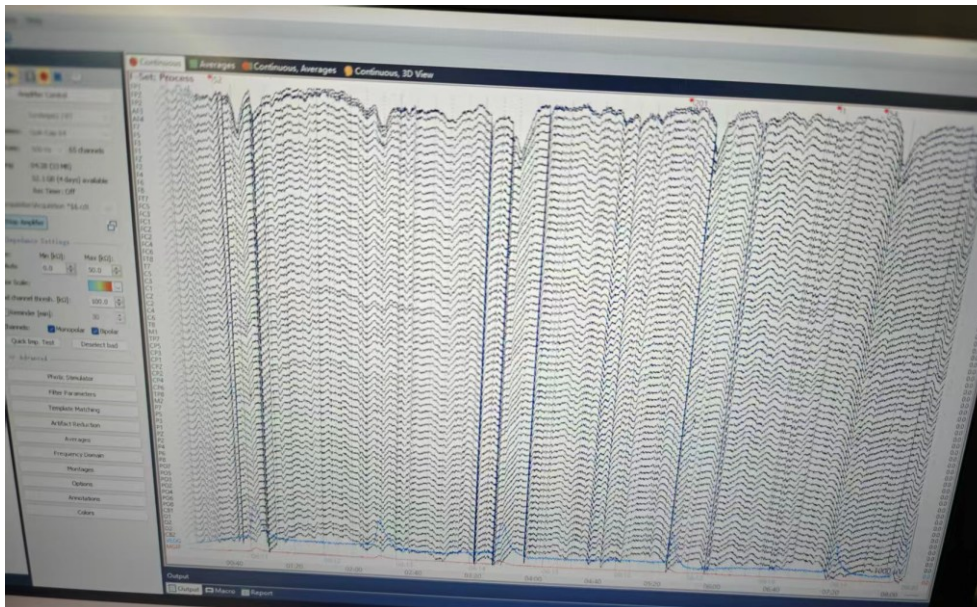


Figure 8. Example of abrupt, widespread voltage transients affecting multiple EEG channels, possibly caused by reference-electrode disturbance or abrupt changes in electrode contact.

2.7 Slow baseline drift

Slow baseline drift appears as a gradual, low-frequency shift in the EEG signal over long epochs. It may affect one or multiple channels and can sometimes produce large, slowly changing voltage deflections.

Possible causes:

- **Sweating:** Changes in skin potentials and the electrode–skin interface caused by sweating can produce very slow voltage fluctuations, often referred to as **sweat artifacts**. If sweating occurs during recording, reduce the room temperature or improve ventilation where possible.
- **Slow head or body movements:** Gradual movement may alter electrode contact or cause mechanical displacement of the electrodes and cables, resulting in slow signal shifts.
- **Changes in electrode–skin contact:** Gradual changes in electrode impedance, loose electrodes, or drying conductive gel can contribute to baseline instability.
- **Long recording sessions:** Prolonged recordings may increase the likelihood of sweating, electrode drying, impedance changes, fatigue, and movement, thereby increasing baseline drift.
- **Participant discomfort or tension:** Discomfort or nervousness may indirectly increase baseline drift by promoting sweating or frequent small movements.

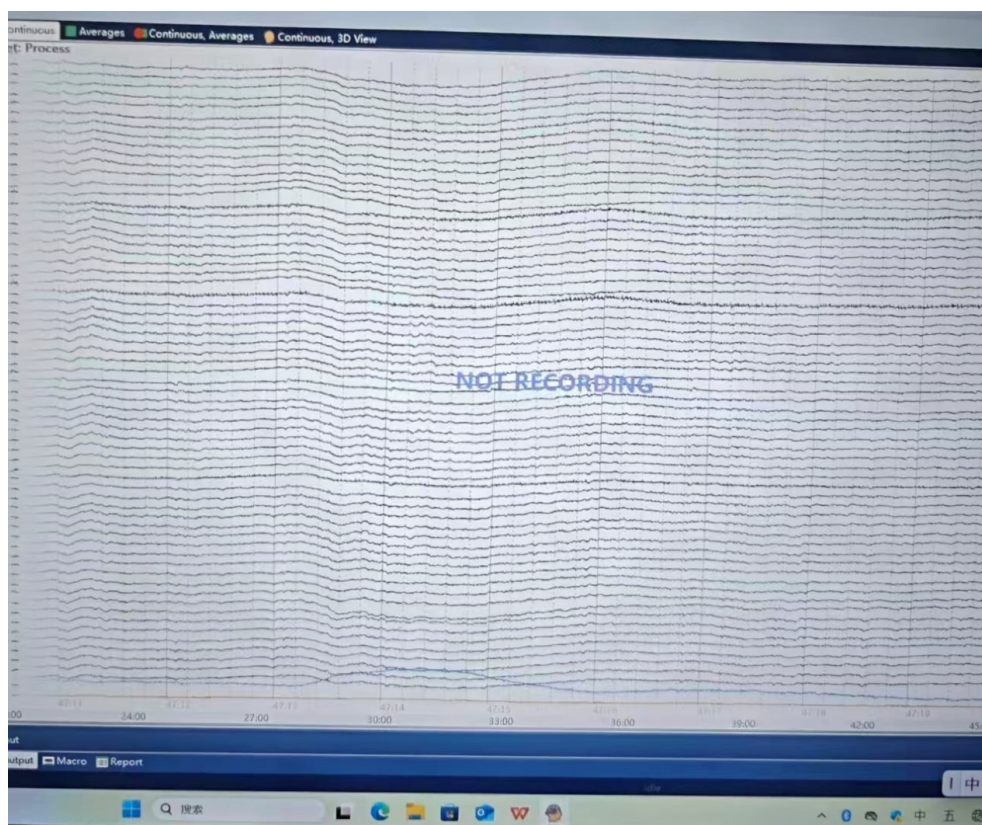


Figure 9. Example of slow baseline drift

Resources

[Artifacts on EEG — Learning EEG](#)

[How to Read an EEG: Waveforms, Montages & Artifacts - ScienceInsights](#)

[Appendix 4. Common Artifacts During EEG Recording - Electroencephalography \(EEG\): An Introductory Text and Atlas of Normal and Abnormal Findings in Adults, Children, and Infants - NCBI Bookshelf](#)

3. Identifying Artifactual Independent Components

3.1 Evaluate multiple views together

Independent components should not be classified from a single feature. Evaluate four complementary views: **scalp topography**, **component time series**, **power spectrum**, and **ERP-image**. ICLabel probabilities are useful decision support but should be reviewed against these features, especially for ambiguous components.

3.2 Brain-related components

- Scalp map: smooth and physiologically plausible rather than dominated by a single electrode.
- Time series/ERP-image: activity is distributed across time and trials rather than consisting only of stereotyped artifact bursts.
- Spectrum: a broadly decreasing 1/f-like spectrum; a posterior alpha-band peak around 8-13 Hz may appear but is not required.
- Decision: retain unless there is clear evidence that the component is predominantly artifactual.

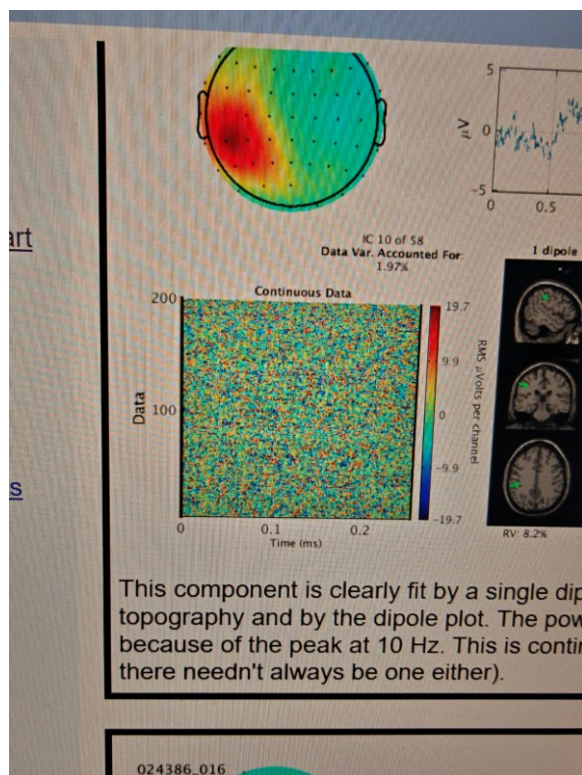


Figure 10. Brain-like scalp projection and distributed activity.

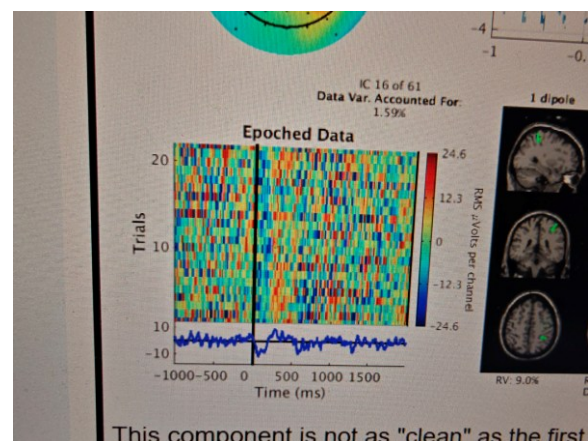


Figure 11. ERP-image example from a likely brain-related IC.

*Illustrative screenshots from the ICLabel training material included in the source document.

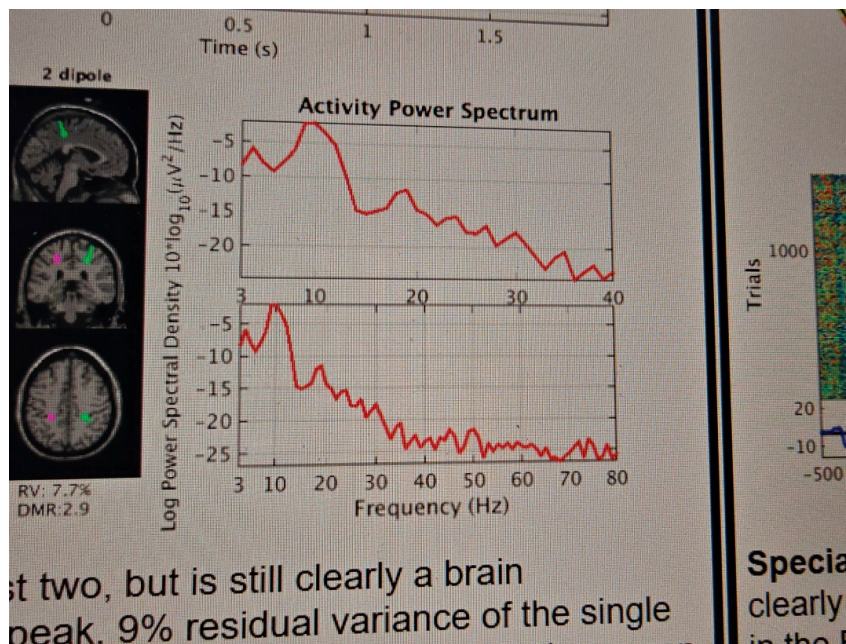


Figure 12. Example power spectra with a plausible low-frequency profile and an alpha-range feature.

*Illustrative screenshot from the ICLabel training material included in the source document.

3.3 Ocular components

- Scalp map: strong frontal projection; blink components are often bilaterally frontal, whereas horizontal eye-movement components may show a left-right polarity gradient.
- Time series: large, stereotyped blink or saccade deflections.
- Spectrum: most power at low frequencies with a smooth decline at higher frequencies.
- ERP-image: repeated bands or patches corresponding to individual ocular events across trials.

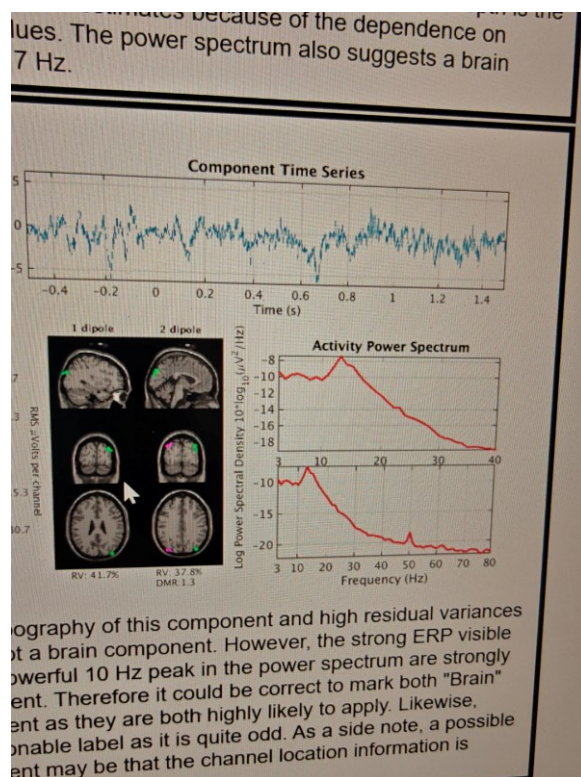


Figure 13. Low-frequency ocular-like component properties.

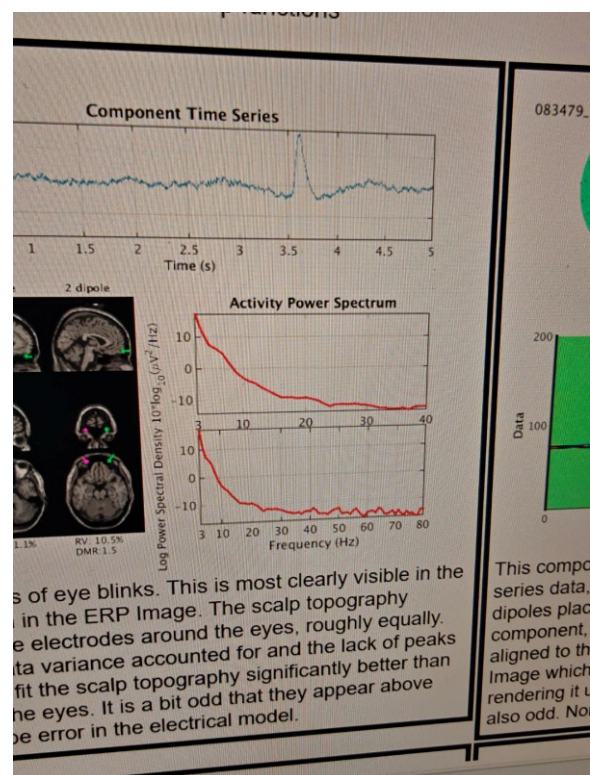


Figure 14. Blink-related time-series and spectral pattern.

*Illustrative screenshots from the ICLabel training material included in the source document.

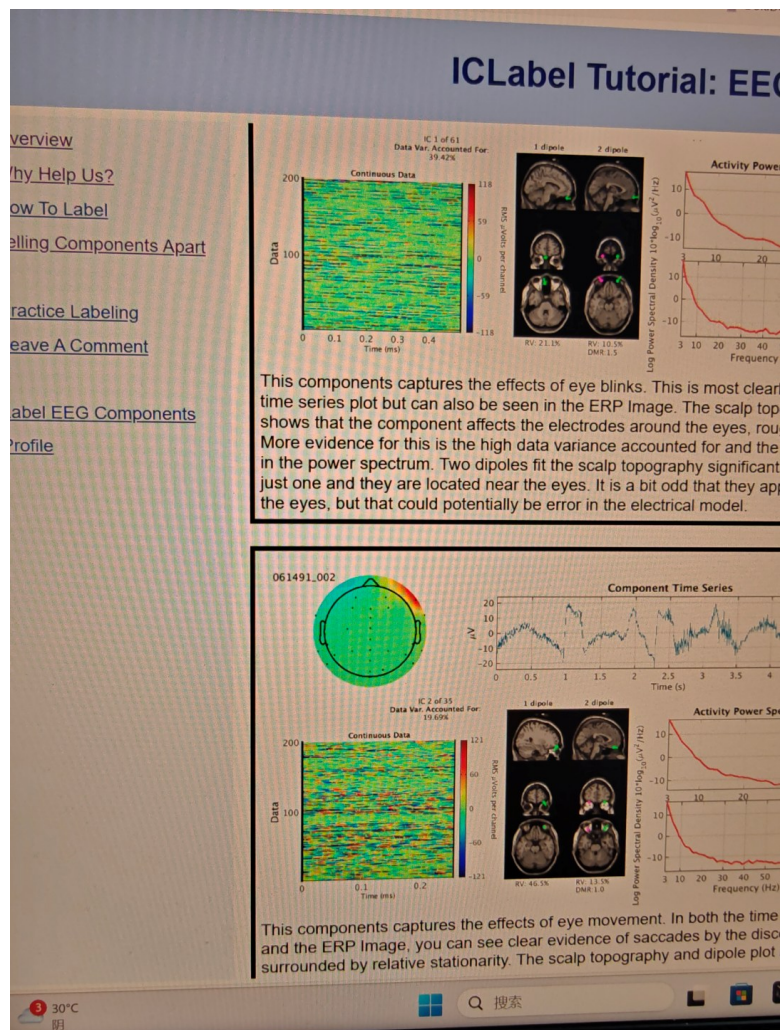


Figure 15. Additional ocular-component examples from the ICLabel tutorial.

*Illustrative screenshot from the ICLabel training material included in the source document.

3.4 Muscle components

- Scalp map: spatially focal, often near the scalp edge, temporal area, jaw, forehead, or neck.
- Spectrum: elevated broadband high-frequency power, commonly above 20 Hz.
- Time series: irregular bursts rather than smooth rhythmic activity.
- ERP-image: fine-grained, speckled, or trial-specific horizontal bands reflecting nonstationary bursts.

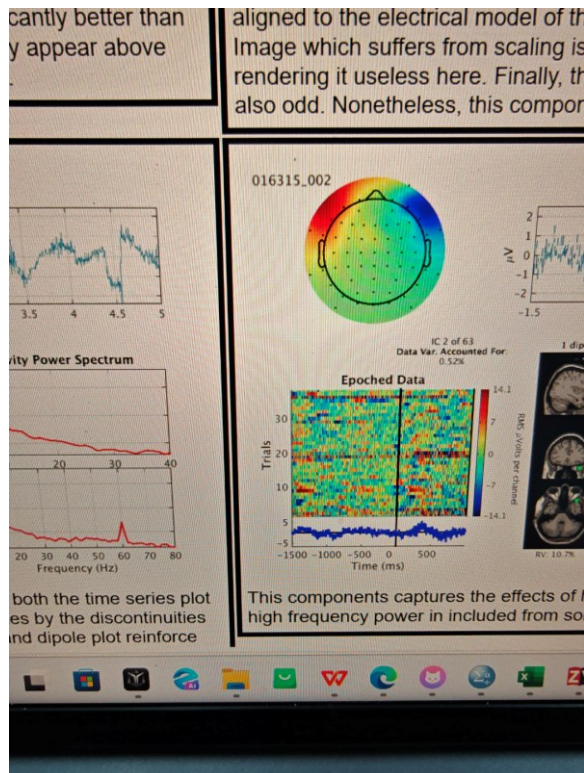


Figure 16. Example of a focal, high-frequency muscle component.

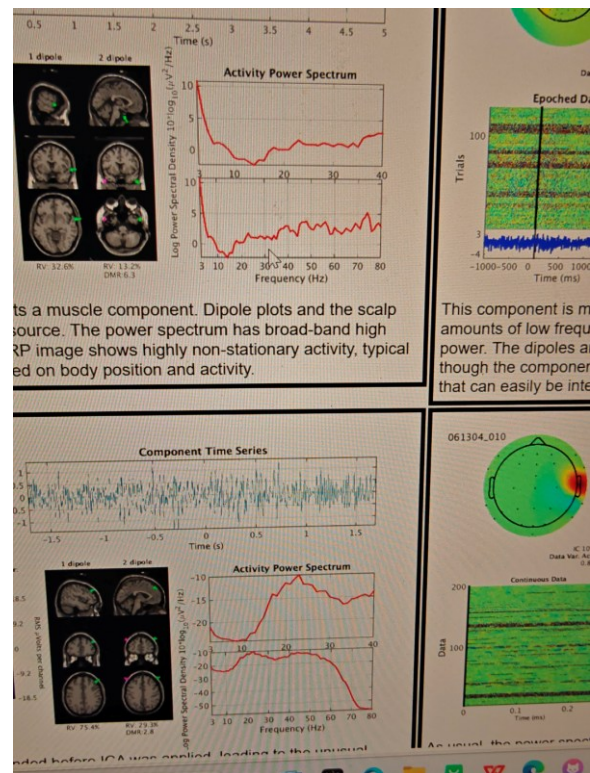


Figure 17. Additional muscle-component examples.

Illustrative screenshots from the ICLabel training material included in the source document.

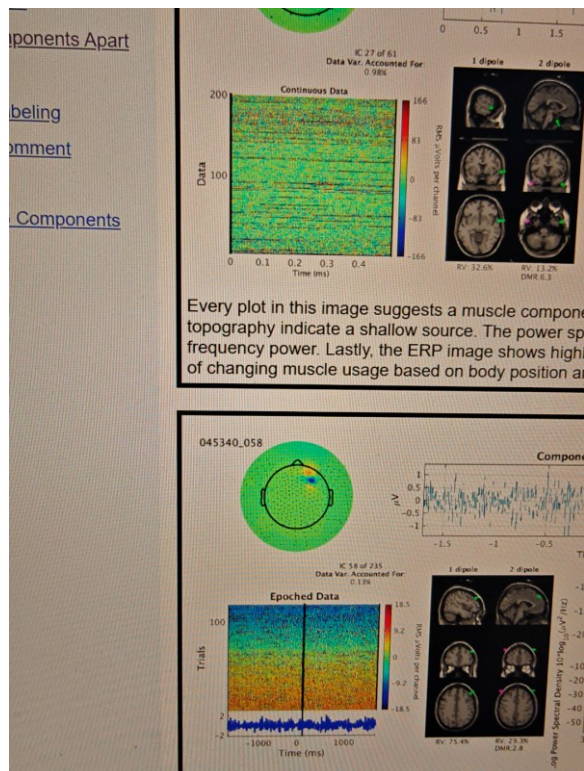


Figure 18. Broad high-frequency power and nonstationary trial structure.

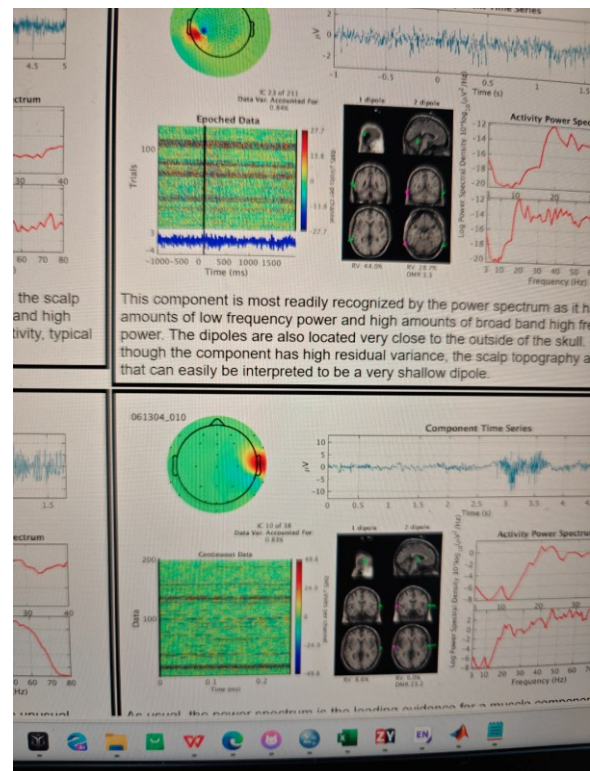


Figure 19. Muscle-related ERP-image patterns.

*Illustrative screenshots from the ICLabel training material included in the source document.

3.5 Cardiac components

- Time series: recurrent sharp complexes at the participant's cardiac rhythm, often roughly 1-2 Hz, with possible harmonics.
- Scalp map: a broad linear gradient rather than a focal cortical-looking projection.
- Confirmation: compare with an ECG channel when available; periodicity is the most important cue.

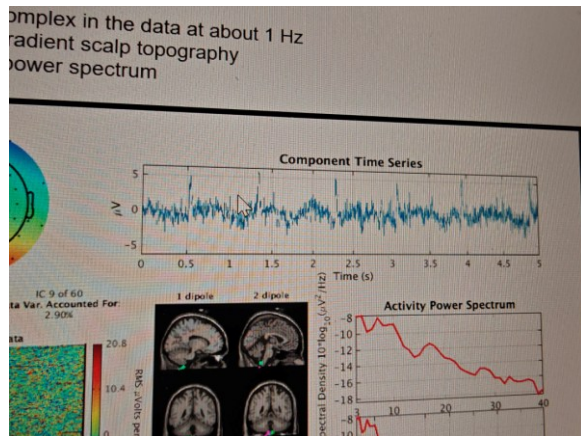


Figure 20. Periodic cardiac-like component activity.

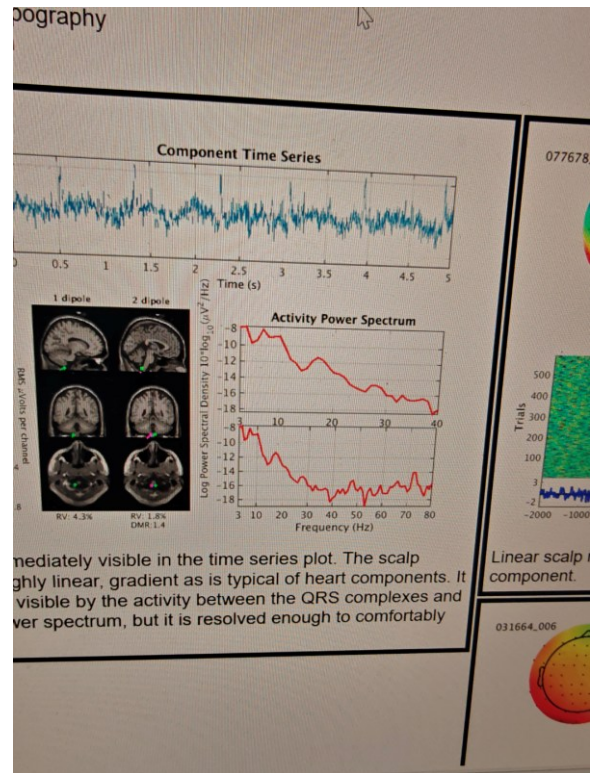


Figure 21. Cardiac component with recurring complexes and a broad scalp gradient.

*Illustrative screenshots from the ICLabel training material included in the source document.

3.6 Line-noise components

Line-noise components typically show a narrow spectral peak at the local mains frequency (50 or 60 Hz) and sometimes its harmonics. The scalp map may be diffuse or localized depending on the source. Confirm the peak against the channel-level spectrum before rejection. A notch filter or dedicated line-noise method should not be used automatically when the spectrum is already clean.

3.7 Channel-noise components

- Scalp map: overwhelmingly dominated by one electrode or a very small local cluster.
- Spectrum/time series: abnormal broadband noise, repeated electrode pops, or nonphysiological fluctuations.
- Interpretation: often indicates that a bad channel remained in the ICA input. Revisit channel cleaning if many such components are present.

Independent Component Labeling

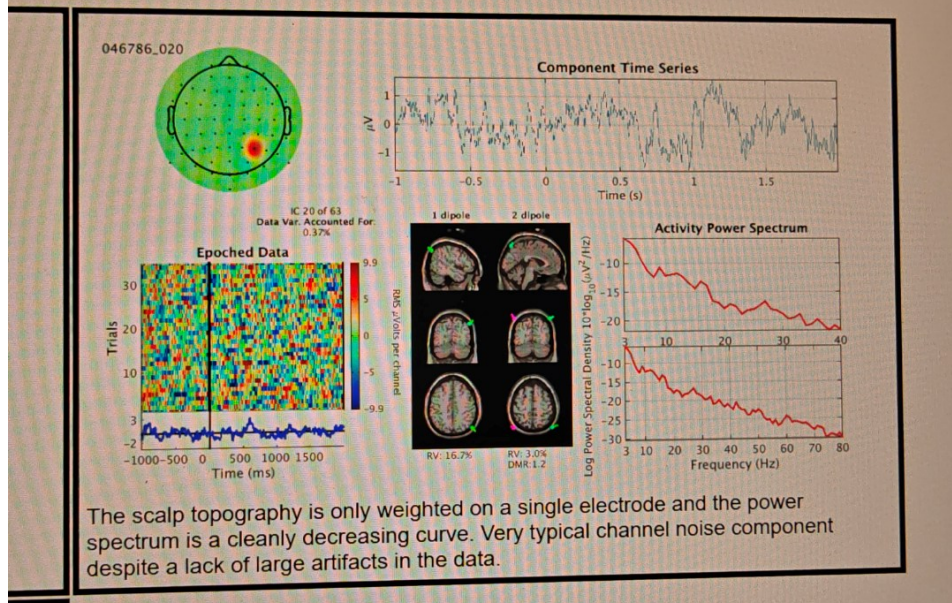


Figure 22. Example of a component dominated by a highly focal, channel-like source.

*Illustrative screenshot from the ICLabel training material included in the source document.

Resources

[SCCN: Independent Component Labeling](#)

4. Special Processing: ICA-Weight Transfer

4.1 Rationale for using separate ERP-analysis and ICA-training branches

ICA decomposition is often improved when slow drifts and other large low-frequency fluctuations are attenuated before training. Consequently, a relatively high high-pass cutoff, such as 1 Hz, is commonly used for the ICA-training dataset. However, this cutoff may not be appropriate for the final ERP analysis because high-pass filtering can attenuate or distort slow ERP activity.

A practical solution is therefore to derive two datasets from the same continuous recording:

- an ERP-analysis dataset using the project-defined low-frequency cutoff, such as 0.1 Hz; and
- an ICA-training dataset using a higher high-pass cutoff, such as 1 Hz.

The values of 0.1 Hz and 1 Hz are examples rather than universal defaults. The final filter settings should be justified with reference to the ERP components, time windows, and experimental contrasts of interest. The higher-cutoff ICA-training dataset is used to estimate the spatial decomposition and should not automatically replace the minimally filtered dataset for ERP measurement.

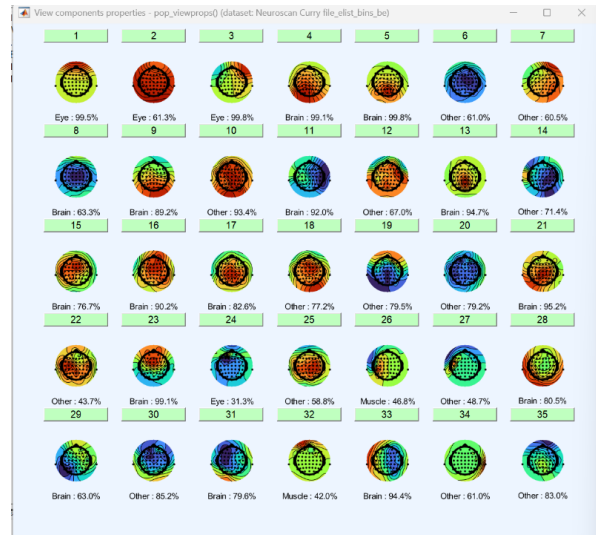
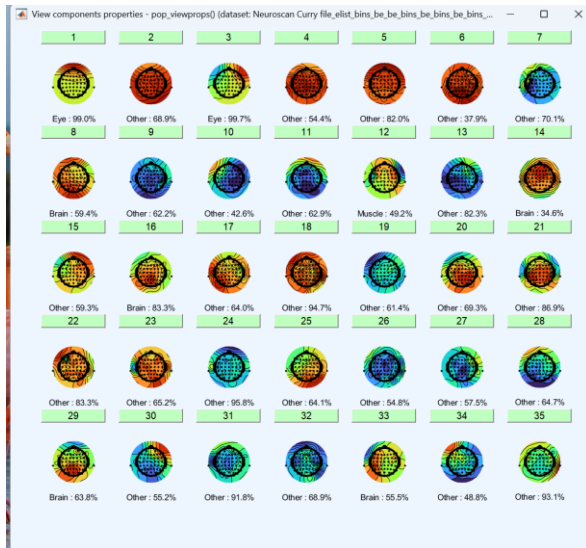


Figure 23. Component overview from the 0.1-Hz filtered ERP branch. Figure 24. Component overview from the 1-Hz ICA-training branch.

4.2 Requirements for valid transfer across branches

The ICA decomposition represents a set of spatial filters operating across channels. The two branches do not need to contain identical time samples, but they must preserve a compatible spatial channel model.

The following properties must match:

- The EEG channels used for ICA must have identical labels and identical order.
- The channel polarity, amplitude units, and channel scaling must be unchanged.
- The reference used when applying the ICA weights must be the same as the reference used during ICA training.
- The effective data rank and the number of ICA dimensions must be compatible.
- The same channel-location information and ICA-channel mapping must be retained.

- Any auxiliary channels excluded from ICA, such as EOG, ECG, or external reference channels, must be handled consistently.
- No spatial transformation that changes the channel mixture may be introduced between ICA training and weight application.

The time samples do not need to be identical. The ICA-training branch may exclude additional non-stereotyped or one-of-a-kind artifacts, including electrode pops, amplifier saturation, recording discontinuities, or gross movement artifacts. The resulting ICA spatial filters may still be applied to the corresponding complete or minimally cleaned recording.

Bad EEG channels should normally be removed from both branches before ICA and interpolated only after artifactual components have been subtracted. Interpolating channels in only one branch before weight transfer may change the spatial structure and data rank.

4.3 Recommended processing sequence

1. Perform the preprocessing operations that must be common to both branches, including channel-location assignment, event checking, identification and removal of bad EEG channels, and selection of the reference used for ICA.
2. Save the minimally filtered dataset as the ERP-analysis branch.
3. Create a duplicate ICA-training branch and apply the project-defined ICA-training high-pass filter.
4. Remove bad segments from the ICA-training branch. These additional segments do not have to be removed from the ERP-analysis branch at this stage.
5. Estimate the numerical rank of the ICA-training data and run ICA using an appropriate number of dimensions. Record the ICA algorithm, parameters, software versions, data rank, number of retained channels, and amount of training data.
6. Inspect the decomposition using component scalp maps, activations, power spectra, and other relevant features. ICLabel may be used as decision support but should not replace visual and physiological evaluation.
7. Copy icaweights, icasphere and icachansind information to the ERP-analysis branch. Component activations should be cleared and recomputed from the ERP-analysis data rather than copied from the ICA-training dataset.
8. Confirm that the transferred component scalp maps and component numbering are preserved. Inspect the component activations in the ERP-analysis branch to ensure that the selected components still capture the expected artifacts.
9. Subtract the selected artifactual components from the ERP-analysis dataset.
10. Interpolate previously removed bad channels using the original channel montage. Continue with epoching, baseline correction, epoch-level artifact detection, and ERP averaging according to the project protocol.

4.4 Use of ICLabel across branches

The manual should specify on which branch ICLabel is run. ICLabel uses information including component scalp topography, power spectrum, and temporal characteristics. Consequently, ICLabel probabilities may change when the same ICA weights are evaluated on datasets with different filtering or retained time samples.

A defensible laboratory convention is to:

- run ICLabel on the ICA-training branch immediately after decomposition;
- record the ICLabel version and probability vector for every component;

- use the probabilities as decision support rather than an automatic rejection rule;
- record the final rejected component indices and the manual justification; and
- visually confirm the selected components after transferring the decomposition to the ERP-analysis branch.

Alternatively, ICLabel may be run on the ERP-analysis branch, but the laboratory must use this choice consistently and report it explicitly. ICLabel results from different branches should not be treated as interchangeable.

5. Reference

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6. Online Resources

EEGLAB preprocessing tutorials: https://eeglab.org/tutorials/05_Preprocess/

EEGLAB artifact rejection and ICA tutorial: https://eeglab.org/tutorials/06_RejectArtifacts/RunICA.html

ERPLAB artifact-detection tutorial: <https://github-wiki-see.page/m/ucdavis/erplab/wiki/Artifact-Detection%3A-Tutorial>

EEGLAB discussion mailing list: <https://sccn.ucsd.edu/eeglab/eeglabmail.html>